Experience-aware staffing for Float

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Smart assign suggests who should fill an open role by matching job titles and checking who is free; per Float's help article it doesn't look at what people have worked on. I built a version that also uses past work (same client, similar projects, matching skills, familiar teammates) and tested both on an agency loaded into Float. This note shows how the test works and what it can and cannot tell you. The code is in the repo.

How I tested it

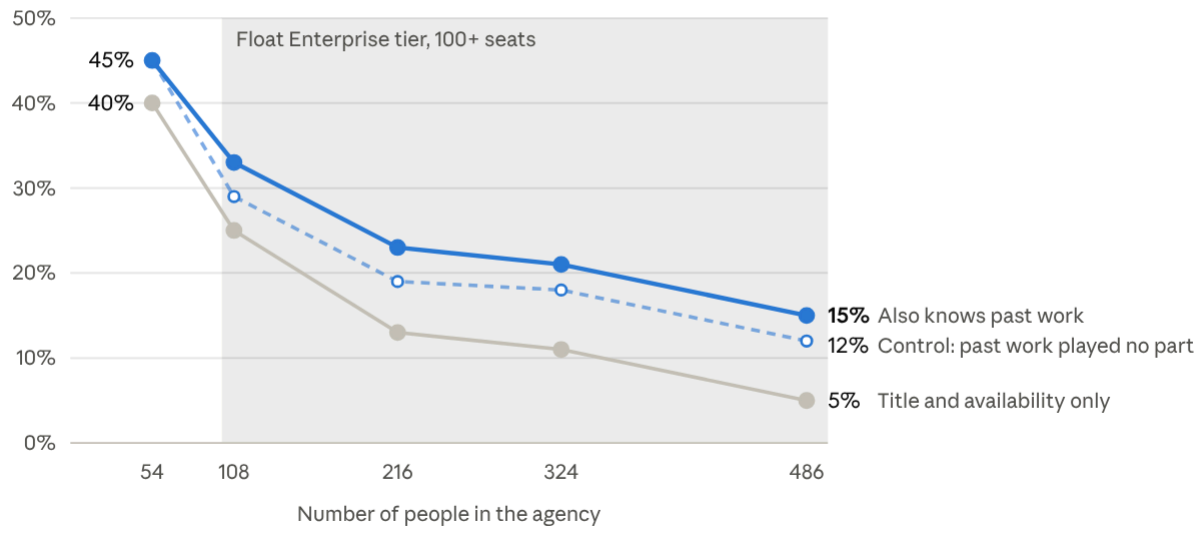
There is no public Float data, so I made some. I invented a 54-person agency with 18 months of projects and 459 staffing decisions, and loaded it into a Float trial through the public API (about 960 calls, within Float's rate limits, resumable if interrupted), so Float's own Smart assign could run on it. Everything downstream reads it back out through the same API, read-only. A dbt project rebuilds the inputs in SQL with tests, including one that no input uses anything that happened after the decision it feeds.

The test is simple. Take each past staffing decision, hide who got the job, and ask both versions to suggest someone using only what was known at the time. Then check: was the first suggestion the person who actually got the job? The chart shows how often it was.

To see what happens as teams grow, I generated bigger agencies the same way, up to 486 people, and ran the same test on them outside Float.

Knowing people's past work helps once a team passes about 100 people

How often the first suggestion was the person who actually got the job



simulated agencies of each size, 3 versions each, averaged

What the chart shows

In a 50-person agency there are usually only two or three Senior Designers, so "a free Senior Designer" is nearly the answer already and the two lines sit together. In a 300-person agency there might be fifteen, and knowing who has done this kind of work starts to separate them. That is the gap between the grey and solid blue lines.

Why there is a dashed line. Past work matters in this agency because I made it matter when I invented the staffing history. To check the model is not just finding what I planted, I also built agencies where past work played no part in who got staffed. On those, the model learns that history carries no signal and gives it almost no weight, so the blue line falls back to the dashed one. The honest size of the effect is the distance between the two blue lines: 3 to 5 points from about 100 people up, nothing at 54.

What this does not show

The agency is invented and the staffing rules are mine. I staffed it with a simple rule that favoured people who had worked for the client before, had done similar work, got on with the producer and were free, plus a dose of randomness and some "whoever is free" picks. So the result says how the two approaches behave as teams grow, not what Float's customers would see. The real test is running the same thing on real staffing history. It needs nothing exotic: allocations, project roles and time off, which are already in your warehouse.